SC

13/08/26 → PDT

18/08/26

Fuzzy Set

→ Proposed by Zadeh in 1965

→ Crisp / Convention Set

↳ U = universal set

① A list method:

$A = \{a_1, a_2, \ldots, a_n\}$ ⇒ all

② Rule Method

$A = \{x / P(x)\}$ ⇒ according to rule

③ Characteristic function

$$M_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

↓

Belongingness of element x ($x \in U$) in set A.

Eg: $U = \{10, 20, 30, 40, 50\}$

① $A = \{10, 20, 30\}$

② $A = \{x / x < 40\}$

③ $M_A(10) = 1, M_A(20) = 1, M_A(30) = 1$

$M_A(40) = 0, M_A(50) = 0$

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$A = \{(x, \mu_A(x))\} = \{(1_0, 1), (2_0, 1), (3_0, 1), (4_0, 0), (5_0, 0)\}$

In crisp / Convention set $\Rightarrow \mu_A(x) \in \{0, 1\}$

This means (it says) that an element either belongs to a set or it doesn't.

so, now fuzzy sets are to be defined.

↳ It is a set where belongingness value lies in b/w 0 & 1.

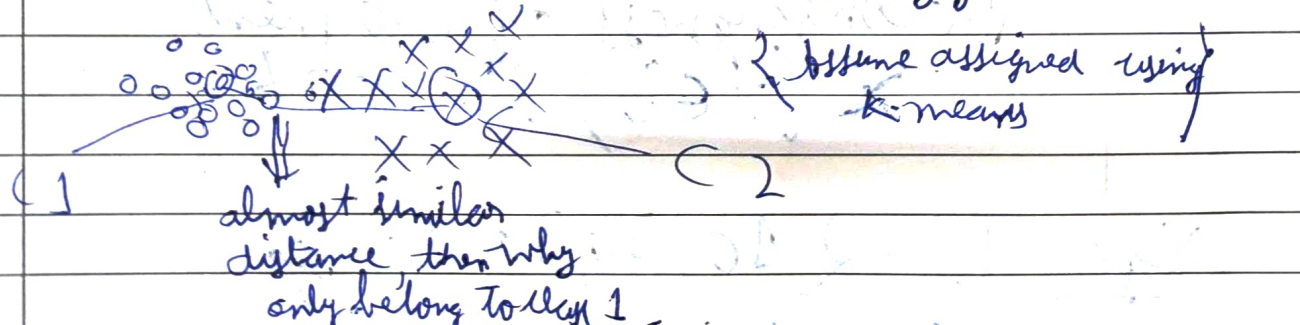
→ Membership f^n or characteristic f^n

$\mu_A(x)$ = Belongingness of element x in set A.

$\mu_A(x) = [0, 1]$

↳ in b/w 0 to 1 (both inclusive).

⇒ Imprecision & uncertainity will be handled with these fuzzy sets.



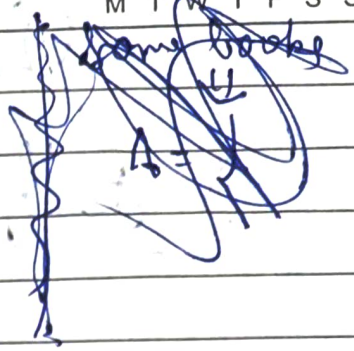
The closer to mean ⇒ more belongingness.

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Define fuzzy set A.

$A = \{(x, \mu_A(x)) / x \in U\}$



Eg: $U = \{0, 1, 2, \ldots, \infty\}$

Crisp sets

young $\rightarrow Y = \{x \mid x \leq 30 \& x \in U\}$

middle aged $M = \{x \mid 30 < x < 60 \& x \in U\}$

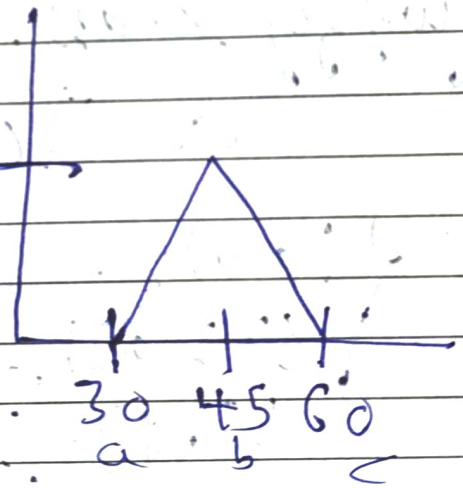
old $O = \{x / x \geq 60 \& x \in U\}$

Fuzzy set

$M =$

$\mu_M(x) \Rightarrow$

Membership for M fuzzy set $\Rightarrow$



$\mu_M(x) =$ triangular (x, a, b, c)

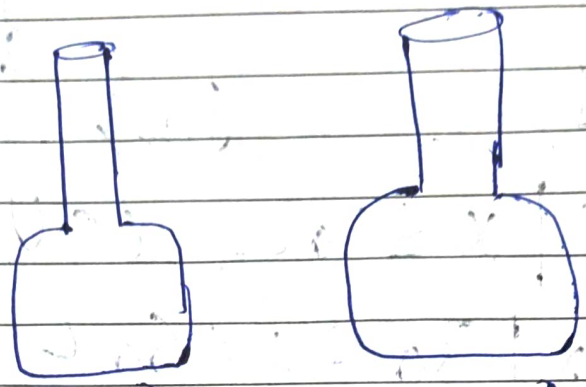
$$\mu_M(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ \frac{c-x}{c-b}, & b \leq x \leq c \end{cases}$$

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Note: ① 'b-a' & 'c-b' need not always be equal.

② a, b, c ⇒ a < b < c

→ Difference b/w M and Probability.



$M_{poison}(\cdot) = 0.5$

↓

50% is poison.

$P(\cdot)_{poison} = 0.5$

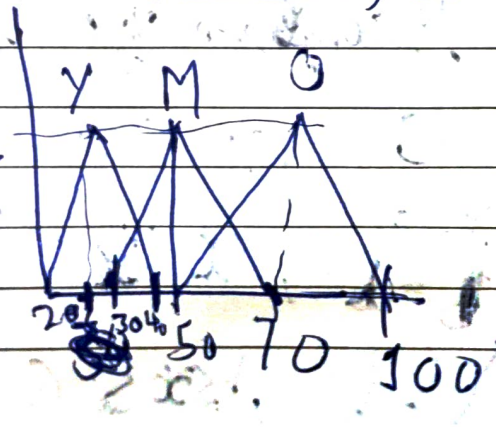
↓

50% chance it contains only poison, 50% it is safe.

Eg: for fuzzy class.

$a = 30,\ b = 50;\ c = 70$

$M = \{(0,0), \ldots, (29,0), (30,0), (31,0.05), (32,0.1), \ldots, (49,0.95), (50,1), (51,0.95), \ldots, (69,0.05), (70,0), \ldots, (100,0)\}$



any can be taken as a, b, c.

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$\mu_y(45)$
$\mu_m(45)$
$\mu_o(45)$ } different belongingness values

for let's say '30' —

$\mu_y(30) = 0.5$
$\mu_m(30) = 0$
$\mu_o(30) = 0$

Note: sum of belongingness over all classes must not exceed 1.

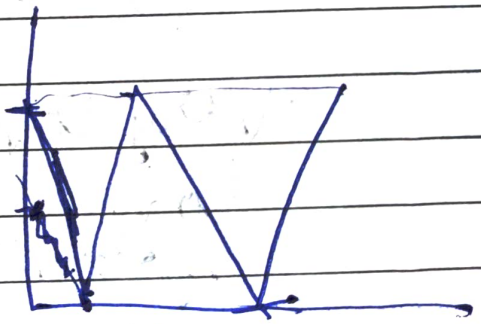
$\sum_{A' \in C} \mu_{A'}(x) \leq 1$ ⇒ for only division into 3 as young, medium, old

⇒ this is not applicable onto adding of 2 different categories (like young and lazy)

Eg: $U = \{0, \ldots, 100\}$

athletic ⇒ μ_A

lazy ⇒ μ_L



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20/08/26 Review

→ Membership function

$M_i(x_i)$ ⇒ belonging ness of pattern x over Class i.

① Triangular function

$$M(x) = \text{tri}(x; a, b, c) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \le x < b \\ \frac{c-x}{c-b}, & b \le x < c \\ 0, & x > c \end{cases}$$

a < b < c

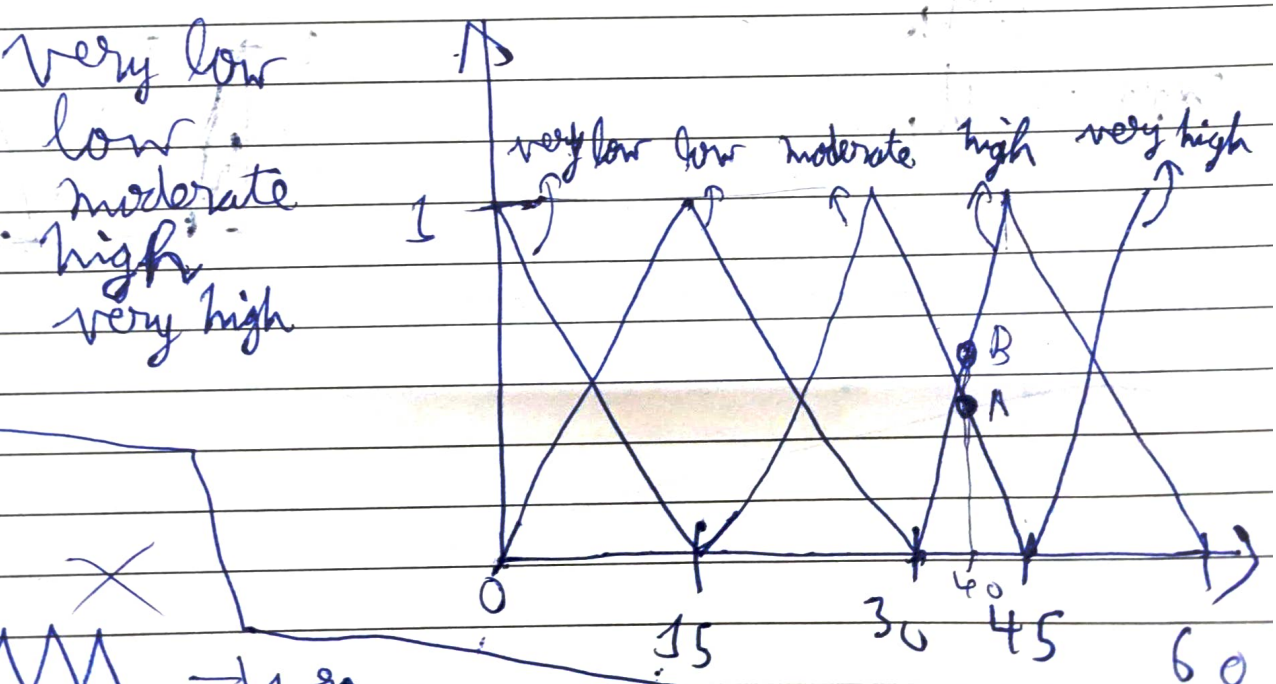
a = b → left open curve

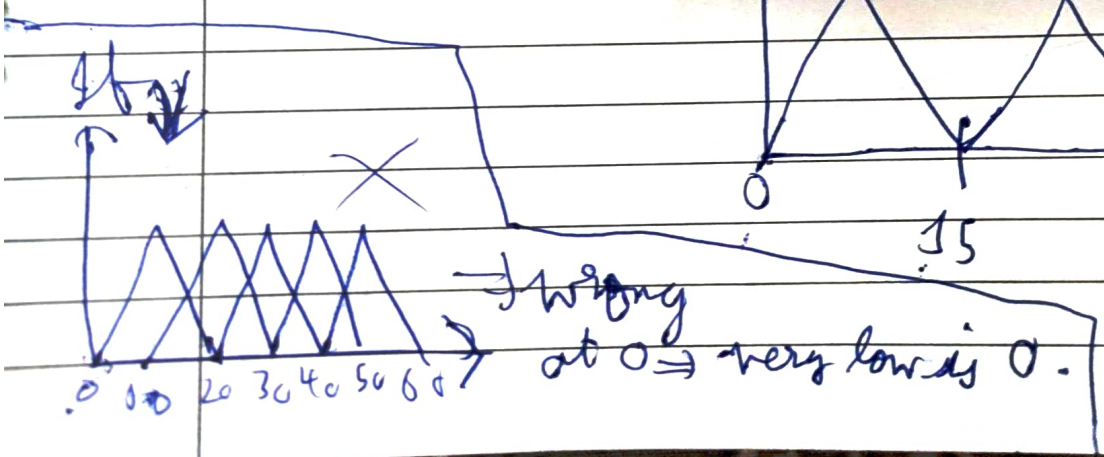
b = c → right open curve

② Temperature

0 to 60

very low
low
moderate
high
very high





⇒ Wrong at 0 ⇒ very low is 0.

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Eg: $\mu_{VL}(40) = 0$

$\mu_L(40) = \frac{45-40}{45-30} = \frac{5}{15} = \frac{1}{3}$

$\mu_M(40) = \frac{45-30}{45-30} = \frac{15}{15} = 1$

$\mu_H(40) = \frac{40-30}{45-30} = \frac{10}{15} = \frac{2}{3}$

$\mu_{VH}(40) = 0$

$\sum_{i \in C} \mu_i(x=40) = 0+0+\frac{1}{3}+1+\frac{2}{3}+0 = \frac{1}{1}$

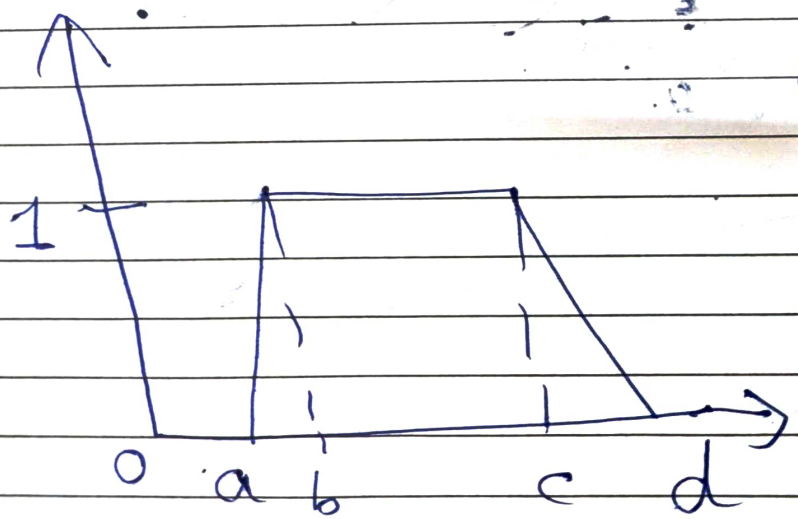
This sum for any class cannot exceed 1 (i.e. ≤ 1)

1

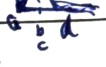
(2) Trapezoidal function

$M(x) = trap(x; a, b, c, d)$

$$\begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & b \leq x \leq c \\ \frac{d-x}{d-c}, & c \leq x \leq d \\ 0, & x > d \end{cases}$$



If $b = c$ ⇓ triangular curve



$a = b = c$ ⇓ left open curve

$b = c = d$ ⇓ right open curve

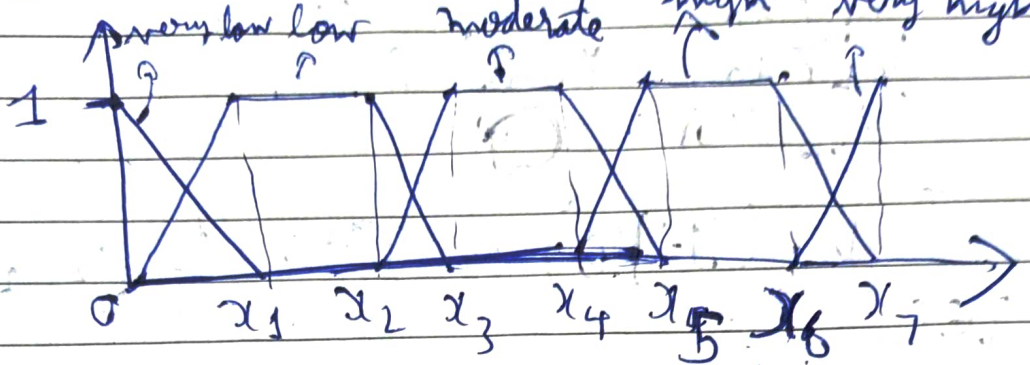
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Let us make trapezoidal function for the temperature



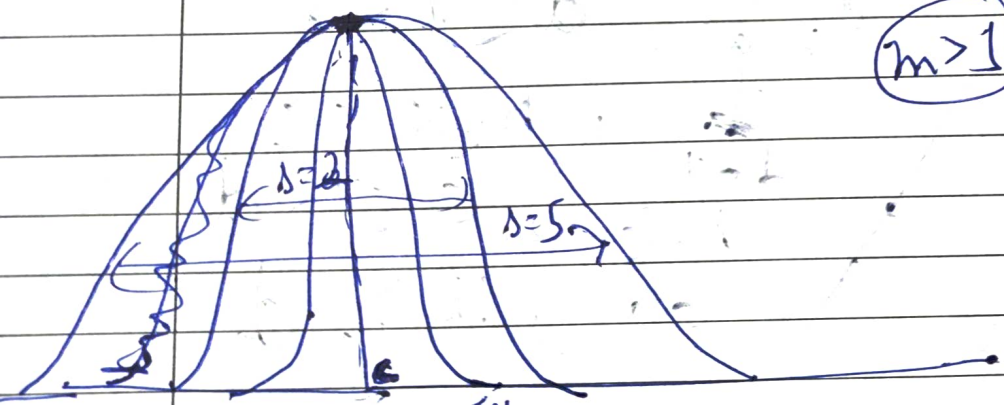
(3) Gaussian membership fn.

$$M(x) = \mu(x; c, s, m) = \exp\left[-\frac{1}{2}\left|\frac{x-c}{s}\right|^m\right]$$

c = centre

s = spread

m = fuzzification parameter



(m>1), generally, m=2.

{ If m=1 then it can be easily shown using crisp set. fuzzy set not needed }

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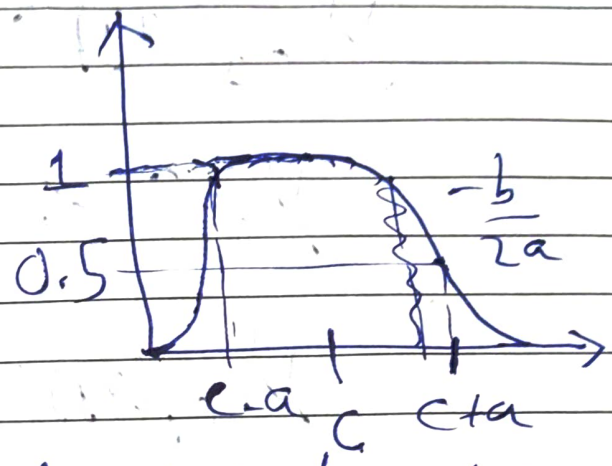
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(4) Bell shaped curve

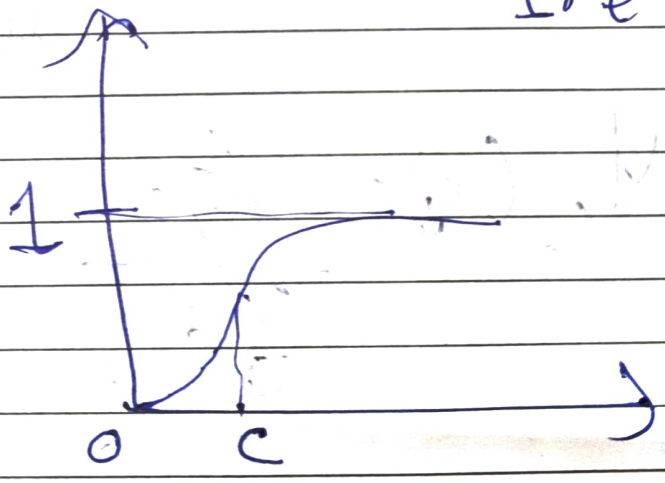
$$M_{bell}(x; a, b, c) = \frac{1}{1 + \left|\frac{x-c}{a}\right|^{2b}}$$

c = centre
a = width
b = slope



(5) Sigmoid Membership function

$$M_{sig}(x; a, c) = \frac{1}{1 + e^{-a(x-c)}}$$



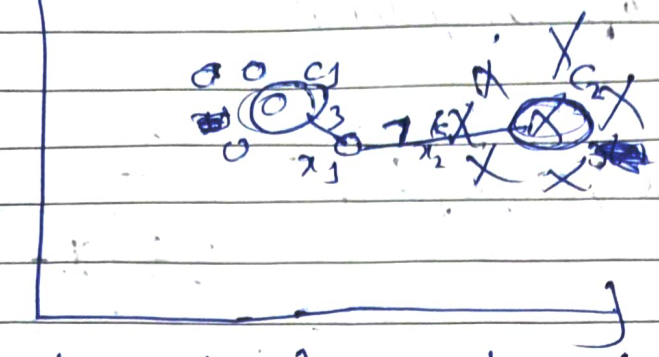
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$C_2 > x > C_1 > 0$



$$\mu_{C_1}(x_1) > \mu_{C_1}(x_2)$$

$$\mu \propto \frac{1}{d}$$

$$\mu_{C_1}(x_1) = \frac{\frac{1}{d(x_1, C_1)}}{\sum_{i=1}^{2} \frac{1}{d(x_1, C_i)}}$$

$$= \frac{\frac{1}{d(x_1, C_1)}}{\frac{1}{d(x_1, C_1)} + \frac{1}{d(x_1, C_2)}} = \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{7}}$$

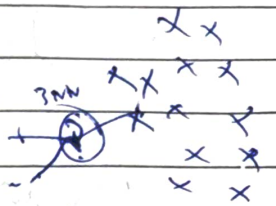
$$\mu_{C_2}(x_1) = \frac{\frac{1}{7}}{\frac{1}{3} + \frac{1}{7}}$$

25/08/26

KNN Classifier

Step 1 → Let there be N patterns; $q_1, q_2, \ldots, q_N$.
And, these patterns belong to C classes,
$w_1, w_2, \ldots, w_C$.

$K \geq 1$, a positive integer.



Step 2: For a test pattern x, calculate K no. of nearest neighbour of x from the training set.

$k_j = 2$
$k_i = 1$
$k_j = 0$
3

[Calculate $d(x_i, x)$ $\forall i = 1$ to N and select K no. of training patterns where $d(\cdot)$ is min^n]

Step 3: Among K Nearest neighbour, K_i belongs to class w_i.

$$\sum_{i=1}^{C} K_i = K$$

Assign x into class w_i if $k_i > k_j$, $\forall i \neq j$

Resolve ties arbitrarily.

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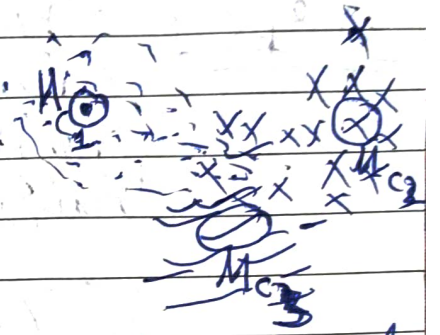
→ But problem arises when there is overlapping information, i.e. no clear boundary b/w klasses.

→ So, Need to use fuzziness.

Fuzzy KNN classifier

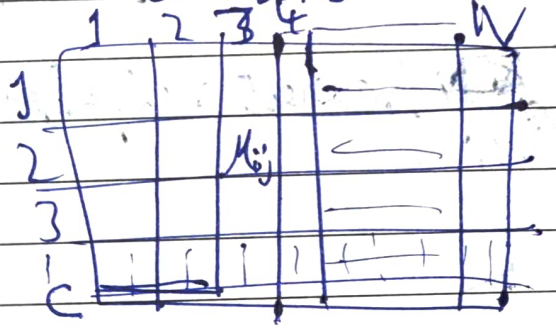
Imprecisely defined class — vagueness of data

⇒ Points do not belong to 1 class only.



Step 1 → same

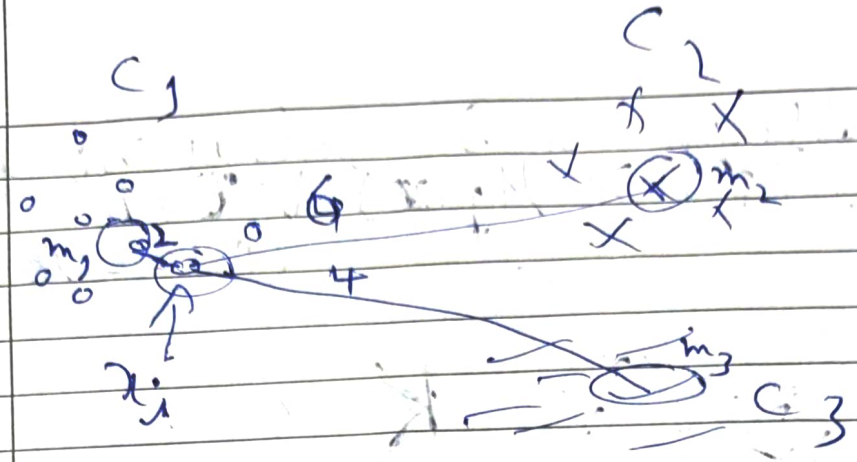
Step 2 → calculate membership Matrix U of dimension C X N ($U_{C \times N}$).



M_{ij} = element of U
= Belongingness of pattern x_j — class w_i

$$M_{ij} = \frac{\left[\frac{1}{d(x_j, m_i)}\right]^{2/m-1}}{\sum_{i=1}^{c} \left[\frac{1}{d(x_j, m_i)}\right]^{2/m-1}}$$

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$$\mu_{ij} = \frac{\frac{1}{d(x_i, m_j)}}{\frac{1}{d(x_i, m_1)} + \frac{1}{d(x_i, m_2)} + \frac{1}{d(x_i, m_3)}}$$

(largest)

$$= \frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{6} + \frac{1}{4}}$$

$$\mu_{2j} = \frac{\frac{1}{6}}{\frac{1}{2} + \frac{1}{6} + \frac{1}{4}}, \quad \mu_{3j} = \frac{\frac{1}{4}}{\frac{1}{2} + \frac{1}{6} + \frac{1}{4}}$$

Step 3: For a test pattern x, calculate k no. of nearest neighbour of x.

Step 4: Calculate

$\mu_i(x)$ = Belongingness of pattern x to class i, $\forall i = 1$ to c

$$= \frac{\sum_{j=1}^{k} \mu_{ij} \left(\frac{1}{\|x - x_j\|^{2/m-1}} \right)}{\sum_{j=1}^{k} \left(\frac{1}{\|x - x_j\|^{2/m-1}} \right)}$$

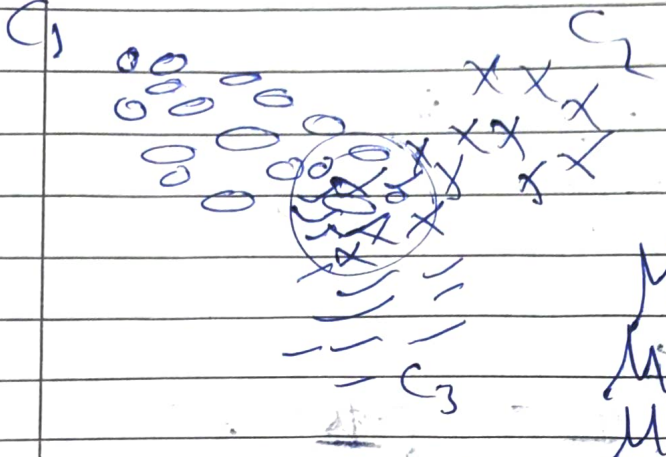
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Step 5: Assign x in class w_i for $M_i(x) > M_k(x)$ $\forall i \neq k$ $\{i, k = 1 \text{ to } C\}$.



$K = 3$

$M_1(x) = 0.31$

$M_2(x) = 0.34$

$M_3(x) = 0.33$

If needed test higher k values like k=5.

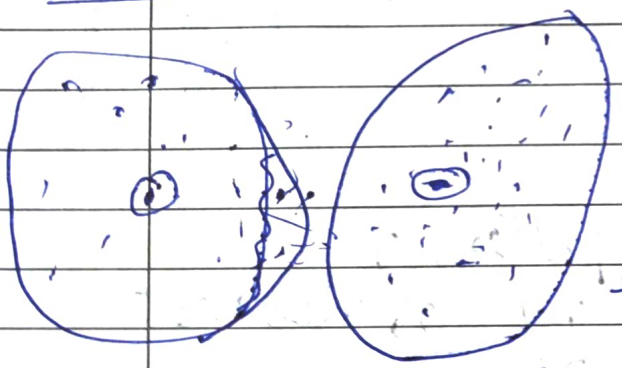
01/09/26

- K → Means Clustering

$k = 2$

Step 1: $S = \{x_1, x_2, \ldots, x_n\} \subseteq \mathbb{R}^p$

K = no. of clusters is given $\{k \geq 2\}$



Step 2: Randomly choose K no. of x, from S and rename them as $y_1, y_2, \ldots, y_K$

Step 3: $C_i = \{x \in S / d(x, y_i) < d(x, y_j)\}$

$\forall i \neq j$; $i, j = 1$ to K

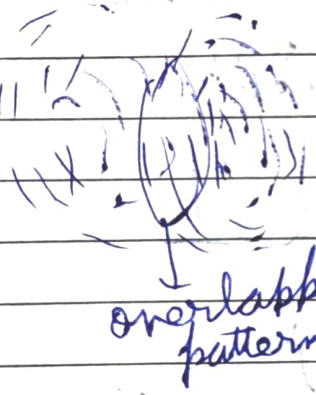
$i = 1$ to K

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Step 4: $z_i = \frac{\sum_{x \in C_i} x}{|C_i|}$; $i = 1$ to k (new mean of cluster i)

Step 5: If $|y_i - z_i| < \in$, $\forall i = 1$ to k
then stop
Else
Rename z_i as y_i, $\forall i = 1$ to k
and return to step 3.

Note: In real data, clusters are almost never seperate from each other, so fuzziness needs to be incorporated.

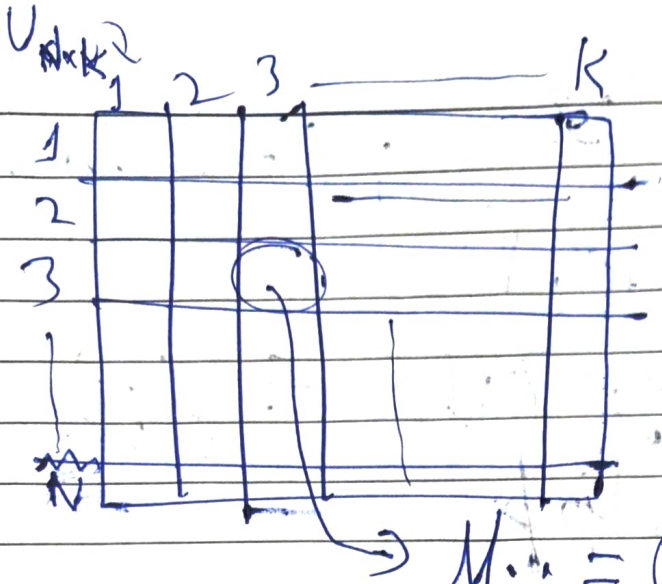


- Fuzzy K-Means Clustering

Step 1 & Step 2 are same.

Step 3: Calculate $U_{N \times k}$, belongingness matrix

N = No. of patterns in dataset
k = No. of clusters



M_{ij} = Belongingness of pattern i (x_i) in class j (C_j).

$$M_{ij} = \frac{\left\| \frac{1}{d(x_i, y_j)} \right\|^{2/(m-1)}}{\sum_{i=1}^{K} \left\| \frac{1}{d(x_i, y_j)} \right\|^{2/(m-1)}}$$

$m \rightarrow$ fuzzy parameter generally $(m=2)$.

So, that $\sum_{j=1}^{K} M_{ij} = 1$

Step 4: $$Z_j = \frac{\sum_{i=1}^{N} M_{ij}^m x_i}{\sum_{i=1}^{c} M_{ij}^m}, \forall j = 1 \text{ to } k$$

Step 5: If $|y_i - z_i| < \epsilon$, $\forall i = 1$ to K stop.

Else rename z_i as y_i and go to step 3.

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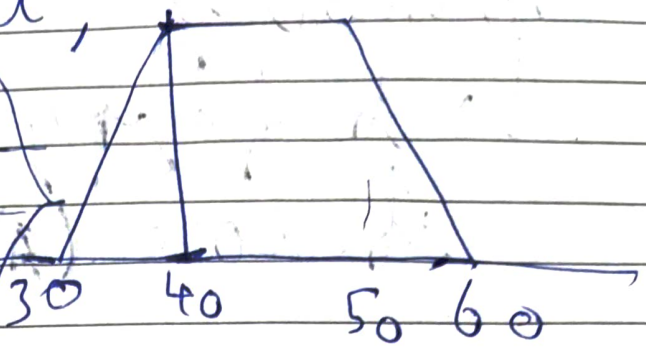
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Important terms

We have seen that,

Middle Age = {(1,0), (2,0), ——, (30,0), (31,0.1), —— (35,0.5), —— (40,1), —— (50,1), —— (55,0.5), ——, (60,0), ——, (100,0)}



$$\text{Support}(A) = \{ (x, \mu_A(x)) \mid x \in U, \ \mu_A(x) > 0 \}$$

↙ contains elements where membership value > 0.

$$\text{Support}(M.A.) = \{(31, 0.1), (32, 0.2), \ldots (59, 0.1)\}$$

$$\text{Core}(A) = \{ (x, \mu_A(x)) \mid x \in U, \ \mu_A(x) = 1 \}$$

↙ contains elements where membership value is 1.

$$\text{Crossover}(A) = \{ (x, \mu_A(x)) \mid x \in U, \ \mu_A(x) = 0.5 \}$$

↙ contains elements where membership value is 0.5

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Normality

↳ A fuzzy set A is normal if $\text{core}(A) \neq \phi$.

α-Cut

$$A_\alpha = \{ x, \mu_A(x) / \mu_A(x) \geq \alpha \}$$

$0 < \alpha < 1$

$$M.A.\ \alpha = 0.3 = \{(33, 0.3), \dots, (57, 0.3)\}$$

⇒ Strong α-Cut

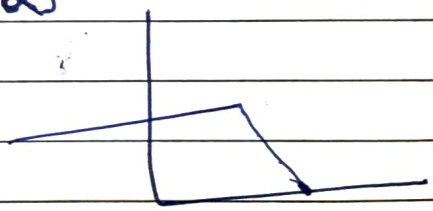
it is making a crisp set.

$$A_{\text{Strong}\alpha} = \{ x / \mu_A(x) > \alpha \}$$

$$M.A_{\text{Strong}\alpha = 0.3} = \{34, 35, \dots, 56\}$$

---

① ⇒ A fuzzy set 'A' is open left if $\lim_{x \to -\infty} \mu_A(x) = 1$ & $\lim_{x \to \infty} \mu_A(x) = 0$



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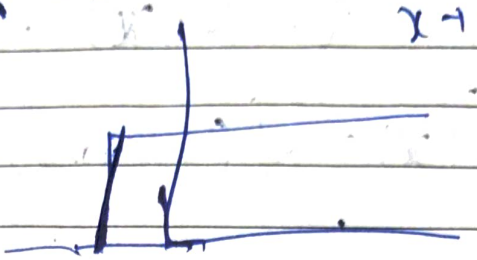
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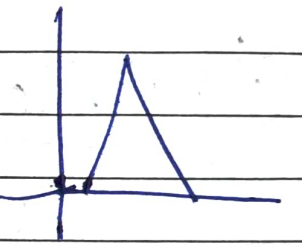
(2) A fuzzy set 'A' is open right

$\lim_{x \to -\infty} \mu_A(x) = 0$ & $\lim_{x \to \infty} \mu_A(a) = 1$



(3) A fuzzy set A is closed

$\lim_{x \to -\infty} \mu_A(x) = 0$ & $\lim_{x \to \infty} \mu_A(x) = 0$



03/09/26

Operation On Fuzzy Set

$A(x) = \{(x_1, 0.7), (x_2, 0.3), (x_3, 0.9), (x_4, 0.1)\}$

$B(x) = \{(x_1, 0.2), (x_2, 0.5), (x_3, 0.7), (x_4, 0.4)\}$

(1) Equality

$A(x), B(x) \in X$ → universal set

$A(x) = B(x)$ iff $\mu_A(x) = \mu_B(x)$ $\forall x$

Quick Work

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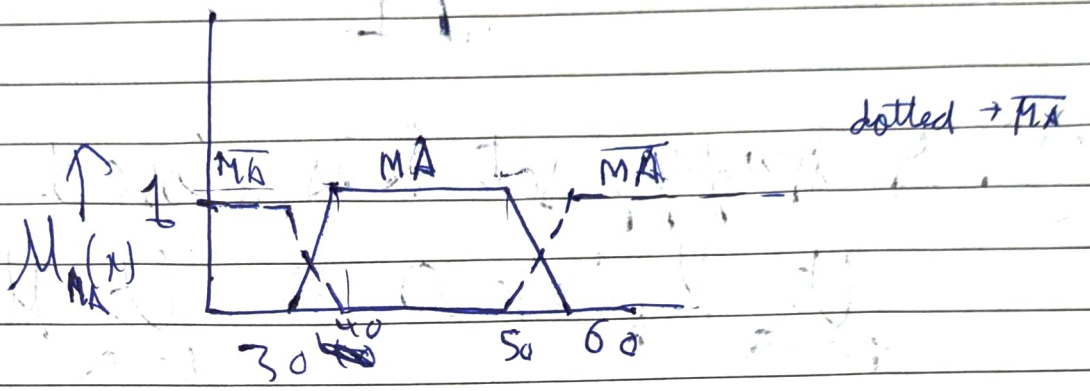
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② Complement

$A(x)$

$\bar{A}(x) = 1 - A(x)$; $\mu_{\bar{A}}(x) = 1 - \mu_A(x)$

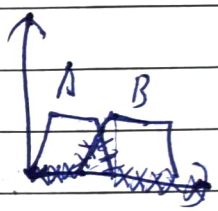
$\bar{A}(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.7), (x_4, 0.9)\}$



③ Intersection

$C(x) = A(x) \cap B(x)$

$C(x) = \{(x, \min(\mu_A(x), \mu_B(x)))\}$



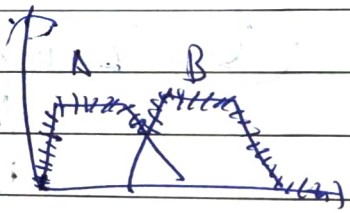
$C(x) = A(x) \cap B(x) = \{(x_1, 0.2), (x_2, 0.5), (x_3, 0.7), (x_4, 0.4)\}$

④ Union

$A(x), B(x)$

$C(x) = A(x) \cup B(x)$

$= \{(x, \max(\mu_A(x), \mu_B(x)))\}$



$C(x) = \{(x_1, 0.7), (x_2, 0.5), (x_3, 0.9), (x_4, 0.4)\}$

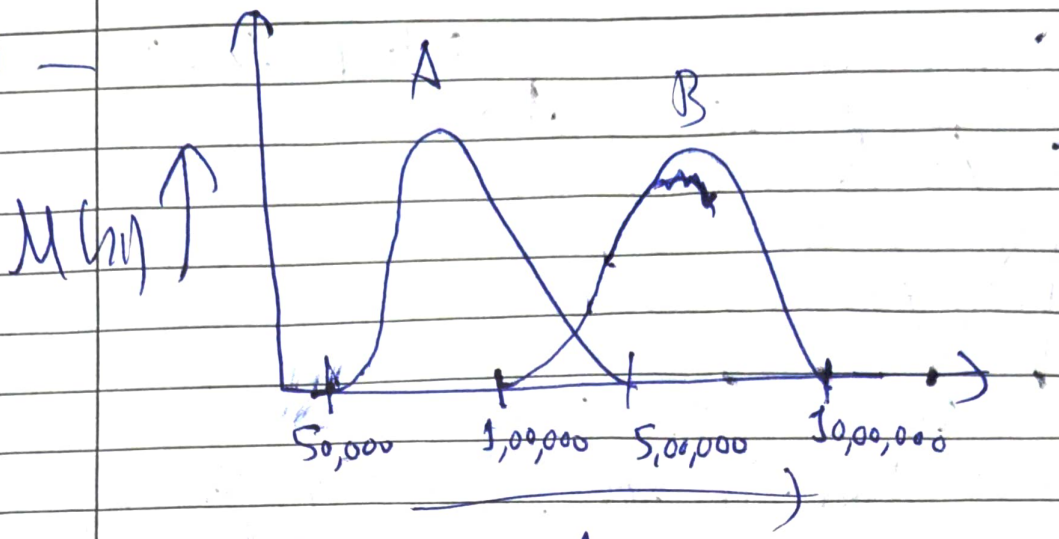
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Q –



A = {Profitable customer}

B = {Large Spender}.

find:

a) all customers deemed profitable or who are large spendor

b) All customers deemed profitable and who are large spendor

c) All customers deemed not profitable

d) " " " not large spendor

e) Persons deemed profitable customers but not large spendor.

draw a table for answer

a) → union of A, B.

b) → intersection

c) → complement of A

d) complement of B.

e) → $A \cap \bar{B}$

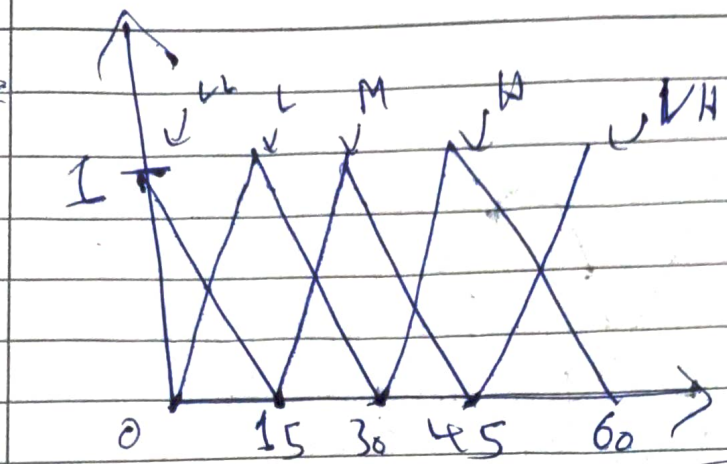
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Eg:



Temperature

If we ask temp. in low as well as medium.

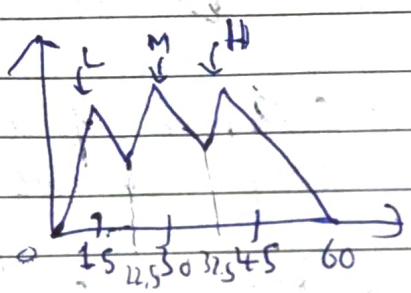
↳ and → intersection

either either/or/or → union

$C = L \cup M \cup H$

Find C in functional form

↳ write membership function.



$$\mu_C(x) = \begin{cases} \frac{x-0}{15-0}, & 0 \le x \le 15 \\ \frac{30-x}{30-15}, & 15 \le x \le 22.5 \\ \frac{x-15}{30-15}, & 22.5 \le x \le 30 \\ \frac{45-x}{45-30}, & 30 \le x \le 37.5 \\ \frac{x-30}{45-30}, & 37.5 \le x \le 45 \\ \frac{60-x}{60-45}, & 45 \le x \le 60 \end{cases}$$

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(5) Algebraic Product

$$C(x) = A(x) \cdot B(x)$$
$$= \{(x, \mu_A(x) \cdot \mu_B(x))\}$$

$$C(x) = \{(x_1, 0.14), (x_2, 0.15), (x_3, 0.63), (x_4, 0.04)\}$$

(6) Multiplication of a Crisp Number

$$C = d \cdot A(x)$$
$$= \{(x, d \cdot \mu_A(x))\} \quad ; 0 \le d \le 1$$

$d = 0.2$

$$C(x) = \{(x_1, 0.14), (x_2, 0.06), (x_3, 0.18), (x_4, 0.01)\}$$

(7) Power of fuzzy set

$$C(x) = A^p(x)$$
$$= \{(x, (\mu_A(x))^p)\}$$

$$C(x) = A^2(x)$$
$$= \{(x_1, 0.49), (x_2, 0.09), (x_3, 0.81), (x_4, 0.01)\}$$

(8) Algebraic Sum

$$C(x) = A(x) + B(x)$$
$$= \{(x, \mu_C(x))\}$$

$$\mu_C(x) = \mu_A(x) + \mu_B(x) - \mu_A(x) \cdot \mu_B(x)$$

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$C(x) = \{(x_1, 0.76), (x_2, 0.65), (x_3, 0.97), (x_4, 0.46)\}$

(9) Bounded Sum ⊕

$$C(x) = A(x) \oplus B(x)$$
$$= \{(x, M_C(x))\}$$
$$M_C(x) = \min(1, M_A(x) + M_B(x))$$
$$C(x) = A(x) \oplus B(x) = \{(x_1, 0.9), (x_2, 0.8), (x_3, 1), (x_4, 0.5)\}$$

(10) Algebraic Difference

$$C(x) = A(x) - B(x)$$
$$= \{(x, M_C(x))\}$$
$$M_C(x) = M_{A \cap B^c}(x)$$
$$= M_A(x) \cap M_{B^c}(x)$$
$$C(x) = A(x) - B(x)$$
$$= \{(x, \min(M_A(x), 1 - M_B(x)))\}$$
$$= \{(x_1, 0.7), (x_2, 0.3), (x_3, 0.3), (x_4, 0.7)\}$$

(11) Bounded Difference ⊖

$$C(x) = A(x) \ominus B(x)$$
$$= \{(x, M_C(x))\}$$

$$\boxed{M_C(x) = \max(0, M_A(x) + M_B(x) - 1)}$$

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$C(x) = A(x) \ominus B(x) = \{(x_1, p), (x_2, 6), (x_3, 0.6), (x_4, 0)\}$

→ Properties

(1) Commutativity ⇒ $A \cap B = B \cap A$, $A \cup B = B \cup A$

(2) Associativity ⇒ $A \cap (B \cap C) = (A \cap B) \cap C$, $A \cup (B \cup C) = (A \cup B) \cup C$

(3) Distributivity ⇒ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(4) Idempotence ⇒ $A \cup A = A$, $A \cap A = A$

(5) Identity ⇒ $A \cap \phi = \phi$, $A \cup \phi = A$ | $A \cup X = X$ (X: universal set), $A \cap X = A$

(6) Transitivity ⇒ If $A \subseteq B$, $B \subseteq C$, then $A \subseteq C$

(7) Involution ⇒ $(A^c)^c = A$

(8) De Morgan's law ⇒ $(A \cap B)^c = A^c \cup B^c$, $(A \cup B)^c = A^c \cap B^c$